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Chapter 34: The Historical Selection of Artificial Intelligence Paths—The State, Academia, Capital, and the Alternative Field (1943–2026)

Abstract

The currently popular narrative of AI progress—industry self-reporting, media repetition, investor roadshows—tends to attribute the victory or defeat of a path to algorithmic superiority or the supply of computing power. This is precisely the self-reported discourse on which the Large Language Model (LLM), the current winner of selection, depends. However, serious histories of computing and AI have long documented structural factors such as wartime cryptography, the funding logic of the Advanced Research Projects Agency (ARPA), the funding cut-offs of the AI winters, and the Fifth Generation Computer Systems national project (Russell & Norvig, 2021; Haigh & Ceruzzi, 2021; Feigenbaum & McCorduck, 1983; NRC, 1999). This chapter does not invent these dimensions but classifies the evolution from 1943 (the McCulloch-Pitts neural network model—the direct starting point of the LLM's own algorithmic lineage) to the present into the ebb and flow of four kinds of **selection mechanisms**—national security and bureaucratic selection, academic faction and paradigm selection, capital selection, and the orthogonal selection of the alternative field (finance). These four mechanisms are organized by three relationships: **superposition, substitution, and orthogonality**.

From 1943–1969, state selection dominated; interest rates were not in the selection function. From 1969–1989, academic faction selection was superimposed, causing paths such as statistical pattern recognition, connectionism, and cybernetics to lose influence, be diverted, or be renamed within the mainstream AI narrative for an

extended period. The nineteen years from 1990–2008 serve as the control baseline for capital selection: this period saw several pronounced rises and falls in interest rates, ending in a near-zero rate environment; monetary conditions were not the dominant selection variable for the evolution of AI paths during this period—because cutting-edge AI research had not yet entered the primary selection field of capital. The prolonged low-rate environment from late 2008 to early 2022 allowed the capital selection field to absorb cutting-edge AI at scale for the first time; the selection function was "the credibility of the grand narrative leading to Artificial General Intelligence (AGI)." However, the counterexample of Japan from 1982–2025 defines the boundary conditions—despite the same low interest rates and central bank balance sheet expansion, the multi-path parallel landscape did not emerge due to the alienation of capital channels, the concentrated bets of state capital, and the absorption of low-cost funds by zombie loans. The high interest rate environment from 2022 onward made capital place greater weight on near-term commercial validation; LLMs benefited first because ChatGPT could more readily provide near-term product proof, while some embodied intelligence projects still received large-scale financing during the high-rate period on the strength of deployment intentions and industry-investor endorsement.

A path screened out by one field does not necessarily exit entirely. An approach that loses influence in one evaluative setting need not disappear from every field. Finance provides cases of personnel migration, methodological transfer, and mathematical family resemblance. These connections differ in evidential strength and should not all be treated as direct transmission chains. Together, however, they suggest that survival and opportunities for application depend on the evaluative setting. Lock-in is usually relative to a particular selection field, rather than an ontological termination.

Keywords

selection mechanisms, selection function, capital selection, monetary conditions, commercial viability, silenced paths, alternative field, mathematical family resemblance

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The previous chapter traced how the city's clock is being replaced within the attention economy. This chapter pushes the perspective further upstream: **what is selecting AI paths, and how the selection mechanism itself switches across time periods**.

§1 Introduction: Four Selection Mechanisms and Three Relationships

Selection mechanism refers to the structural forces that determine which technological paths receive resources, are scaled, and enter historical writing. It is not equivalent to "scientific correctness"—many paths later deemed correct were screened out at the time; nor is it equivalent to "market demand"—many paths with demand were suppressed by non-market mechanisms. Each selection mechanism contains a **selection function**: an implicit or explicit criterion used to score paths.

Over eighty-plus years, at least four dominant selection mechanisms have shaped the life and death of AI paths:

- **National Security and Bureaucratic Selection**: The selection function is "does it serve national strategic goals." 1943–1969 was its period of dominance; after the 1970s, it reappeared intermittently in forms such as export controls and industrial policy. In the 2020s, U.S. export controls on advanced lithography equipment to China tightened the R&D constraints on Huawei's semiconductor team, which in

2026 publicly proposed a methodology substituting time scaling for geometric scaling (He, 2026; on the industrial power structure underpinning this selection capability—technological chain chokepoints and financial chain control—see Li, 2022), a contemporary sample of this selection mechanism continuing to the present day.

- **Academic Faction and Paradigm Selection**: The selection function is "does this look like intelligence as defined by this paradigm." Academic paradigms shaped access to research resources and evaluation standards in a phased manner. The resurgence of connectionism in the 1980s, the growing influence of probabilistic and statistical approaches in the 1990s, and the expansion of deep learning after 2012; the times at which different traditions entered the mainstream were not the same.
- **Capital Selection**: The selection function switches with the capital environment—under a zero interest rate environment, it is "the credibility of the grand narrative"; under a high interest rate environment, it places greater weight on near-term commercial validation. Capital selection includes funding, withholding support, and exit decisions concerning investable projects and applications; it does not require an AI label. This chapter focuses on when capital expanded from supporting specific commercial applications to supporting frontier research at greater scale. Multi-path parallelism is one of the possible results—and one of the organizational conditions—of a certain kind of financing environment, not a precondition for the existence of capital selection itself. The shape of capital selection is jointly modulated by four layers of conditions: upstream are monetary conditions (Federal Funds Rate, credit easing, balance sheet expansion pace), which transmit through the capital environment (exit market activity, platform cash flow, M&A windows, risk appetite); the capital environment then acts on specific paths through capital channels (independent VC, bank-affiliated VC, parent company internal capital, government contracts, cloud vendor subsidies) and financing criteria (patience horizon, return requirements, commercial proof window). Interest rates are not capital itself—they are a price signal of the capital environment; capital selection is also not interest rate selection—interest rates change the shape of capital selection but do not directly adjudicate paths. This conceptual hierarchy is repeatedly used in §§5–7; the Japan case in §6 is precisely the clearest proof of this hierarchy from the negative side.
- **Orthogonal Selection of the Alternative Field**: Paths screened out by one field can be adopted in another—an alternative field with markedly different evaluative criteria can provide new room for application for disfavored paths ("orthogonal" is used by analogy, meaning the evaluative criteria barely overlap, not in the mathematical sense of a zero inner product); this chapter calls such re-adoption "orthogonal selection," without treating it as a necessary condition for all cross-field survival. The most systematic sample is the financial field's absorption of mathematical traditions discarded by academic AI—in quantitative investing, risk-adjusted out-of-sample performance is an important evaluative standard, with the Sharpe ratio as one representative measure; pricing, risk management, and asset allocation have their own task-specific requirements.

The first three mechanisms primarily distinguish resource allocation and evaluation systems, while the fourth concerns re-adoption across evaluative settings—these categories can overlap; the periodization compares relative dominance, not the complete presence or absence of any mechanism.

Three relationships exist among these four mechanisms:

- **Superposition**: Multiple mechanisms operate in parallel during the same period. For example, state selection and academic faction selection were superimposed from 1969–1989.

- **Substitution**: a shift in the dominance of resource allocation—which does not mean the earlier mechanism has entirely ceased to operate. For example, after 2009, the zero interest rate environment allowed the capital selection field to absorb "research paths wearing the AI label" itself at scale for the first time (from 1990–2008, the capital field had already effectively intervened in the "subset of statistical learning and pattern recognition stripped of the AI label"; see §4 for details).
- **Orthogonality**: A path screened out by one field is adopted at scale in another. For example, von Neumann complex systems were marginalized in the academic AI field but adopted in the financial field.

Methodological Self-Attribution. The concept of "selection mechanism" used in this chapter follows, in the methodological genealogy, Tang Shiping's Social Evolutionary Paradigm (SEP; Tang, 2013, ch. 1). Tang (2013) divides selection pressures into two sources—the physical environment and humans themselves—and points out that in the ideational dimension of social evolution, "artificial selection rather than natural selection dominates," and that the more direct selection pressure in the ideational dimension is "picking what ideas are to be put into practice or to be backed." AI paths, as products of ideational competition, have their eighty-plus-year selection process structurally located entirely under the source of "humans themselves"; this chapter does not involve the physical selection of hardware evolution by the physical environment—this is an honest demarcation from the upstream hardware layer (transistor scaling, lithography, thermal budget, etc.). The specific contribution of this chapter is to classify the concrete forms of the source "humans themselves" in the history of AI paths into four selection mechanisms and to characterize the superposition, substitution, and orthogonality relationships among them. This continuation also forms a cross-chapter echo with Chapter 3, Assumption One (technological expansion is the product of competitive power selection).

What follows unfolds the ebb and flow of these four mechanisms by time period (§§2–7), then uses one section to specifically address the eighty-plus-year history of the "orthogonal relationship" (§8), and another section to present the evidence left by alternative field migration at the mathematical level (§9).

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§2 1943–1969: The Period of State-Funding Dominance

The first mechanism shaping the life and death of AI paths was national security anxiety.

Wartime and Early Postwar Period (1943–1949): The Purest Form of State Selection. Russell and Norvig (2021, §1.3.1) delimit the birth period of AI as a field to 1943–1956, and almost all the key nodes of this period lie within wartime classified contracts and postwar military funding. In 1943, McCulloch and Pitts published "A Logical Calculus of the Ideas Immanent in Nervous Activity," presenting the first mathematical model of an artificial neuron (McCulloch & Pitts, 1943); Russell and Norvig (2021), in the deep learning chapter, explicitly write: "Deep learning has its origins in early work that tried to model networks of neurons in the brain (McCulloch and Pitts, 1943) with computational circuits"—the LLM, as a product of deep learning, has the direct starting point of its algorithmic lineage right here. Between 1942 and 1948, Wiener organized his work on the statistical treatment of wartime anti-aircraft fire control problems into *Cybernetics* (1948; Haigh & Ceruzzi, 2021, ch. 1). Between 1943 and 1945, at Bletchley Park in the UK, the Bombe designed by Turing and the Colossus built by the team of Newman and Flowers decrypted the German Enigma and Lorenz ciphers—Colossus was not declassified until the 1970s, meaning that the public history of computing/AI missed this line for the first thirty years; even the "history of itself" that the LLM can recount was once systematically screened

by the state's secrecy regime.

On the American side, the ENIAC (1946) at the University of Pennsylvania's Moore School was funded by the Army Ordnance Corps and technically supervised by the Ballistic Research Laboratory (BRL); von Neumann's 1945 EDVAC report was also completed within the framework of a military contract; after the war, the U.S. Navy guided some members of the Communications Supplementary Activity—Washington (CSAW) cryptanalysis group to form Engineering Research Associates (ERA); ERA was acquired by Remington Rand in 1952, and William Norris and others later left to found Control Data Corporation (CDC) (Haigh & Ceruzzi, 2021, ch. 1). In 1949, Hebb proposed the learning rule of synaptic strengthening; in 1950, Turing gave an operable criterion for intelligence in "Computing Machinery and Intelligence"; the 1956 Dartmouth workshop partially separated "Artificial Intelligence" as a term from the neuro-cybernetic tradition (Russell & Norvig, 2021, §1.3.1).

During this period, academic faction selection had not yet become dominant—AI as an independent discipline was still taking shape; capital involvement was also limited (foundation and industry funding existed but was not dominant). **The dominant selection mechanism was the state**: wartime was cryptography and fire control; postwar was Cold War armament and university military contracts.

AI R&D on the American side was initiated amidst Cold War anxiety. The Advanced Research Projects Agency (ARPA, later renamed DARPA), in the 1960s, channeled large amounts of funding to the Massachusetts Institute of Technology (MIT), Stanford University, and Carnegie Mellon University (CMU) under the logic of funding people rather than projects—with no formal peer review and no commitment to specific deliverables (NRC, 1999). The scale and freedom of these funds followed a logic different from that of commercial capital—the driving force was the fear that the Soviet Union would be the first to build a "thinking machine."

On the Soviet side, the computing path was decided by internal bureaucratic factions. MESM (first operational in 1950, officially commissioned in 1951; Kiev, Lebedev) was the first fully electronic stored-program general-purpose computer on the European continent. Glushkov's OGAS was a nationwide economic computing network plan predating ARPANET (the proposal took shape in the 1960s; its decisive review—the Politburo meeting of October 1, 1970—and its downgrade belong to the institutional conflicts of the following period, and are recorded here for continuity), ultimately vetoed by inter-ministerial power struggles (Peters, 2016, ch. 5): Finance Minister Garbuzov, out of departmental interest, prevented OGAS from falling into the hands of the rival Central Statistical Administration, downgrading the proposal from a comprehensive economic management system to a purely technical network. This outcome shows that the selection function was not "is the technology feasible" but "is it advantageous to the power of this department."

From 1943 to 1969, state funding provided a major resource base for the early AI research examined here, while commercial capital had not yet become the principal allocative force. Academic communities, foundations, and firms also helped shape research directions and resources. This period is the "period of state-selection dominance"—the selection function was "is it accepted by the national strategy," and interest rates as a variable were not part of it.

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§3 1969–1989: Academic Faction Selection Superimposed, The First Winter

The second selection mechanism—academic faction and paradigm—was superimposed during this period.

The Mansfield Amendments (1969, 1970) tightened the permissible uses of defense research funding in stages. The 1969 Amendment required defense research to have a "direct or apparent relationship" to specific military functions; the 1970 Amendment further limited the Department of Defense's support for basic research; the tightening of funding constraints coincided in time with some researchers moving to corporate laboratories such as Xerox PARC (NRC, 1999; Haigh & Ceruzzi, 2021). The first AI funding winter was not triggered solely by technical failure—the Mansfield Amendments were one of several parallel factors; the ALPAC report (1966) negating machine translation and Minsky and Papert's *Perceptrons* (1969) striking at connectionism occurred concurrently, with the contraction of funding logic and the collapse of credibility within the field interwoven.

The Institutional Establishment of the IBM-Compatible Path. In 1966–1967, the Soviet Union turned to the IBM/360-compatible solution for its unified computer series—a decision that constrained the resource space of some indigenous development paths without terminating all domestic architectures (the Soviet government formally resolved to develop a Unified System of Computers modeled on the IBM-360); cloning began in 1968, and the first ES EVM went into production in 1971 (Malinovsky, 2010). This decision was part of a deeper institutional failure—Malinovsky documented multiple causes of the Soviet computing lag (including but not limited to): refusal to cooperate with Western Europe on developing fourth-generation machines; forced copying of the IBM-360 while abandoning existing domestic engineering accumulation; splitting the computing industry into three departments in the 1970s, each redundantly copying American hardware and software designs; underestimating the institutional connections between academic science and industry (see Fitzpatrick's editorial introduction and the main text, Part III). From then on, the IBM-compatible system became the dominant direction of Soviet general-purpose computing, constraining indigenous development—but indigenous architectures such as the Elbrus series—developed under Burtsev and others within the research tradition established by Lebedev—did not terminate, continuing through 1991 (Malinovsky, 2010).

Japan entered the game with national will and fell behind. The Ministry of International Trade and Industry (MITI) launched the Fifth Generation Computer Systems (FGCS) project in 1982, betting on parallel logic programming (Prolog), with the main project covering fiscal years 1982–1992 (eleven fiscal years, ending in March 1993) and an investment of approximately 54 billion yen (data per the Information Processing Society of Japan's Fifth Generation Computer project records). The result was a commercial failure—the general-purpose x86/Unix system and its software ecosystem ultimately overwhelmed the FGCS's specialized path. Song's (2016) summary is: after reaching the world frontier, the Japanese government's policies, to a certain extent, hindered the sustained progress of this industry.

The entire golden age of symbolic AI (1980–1988) occurred in a high interest rate environment with the Federal Funds Rate (FFR) between 6%–19%. Japan's Fifth Generation project was launched when FFR was 14.94%; Lenat launched the Cyc project in 1984 when FFR was between 9%–11%. These projects show that higher policy rates did not block long-term AI investment—but the fact that projects obtained funding is not by itself sufficient to establish that the cost of capital had no effect on investment scale and path selection.

But deeper than this is the **form of academic faction selection**. The symbolist paradigm established by the 1956 Dartmouth workshop took "does this look like intelligence" as the implicit selection function. Under this criterion, statistical pattern recognition was regarded as engineering curve-fitting rather than intelligence, long marginalized in mainstream journals and conferences; connectionism repeatedly lost ground in the mainstream after Minsky and Papert's 1969 *Perceptrons* formally demonstrated the representational limitations of the single-layer perceptron, briefly reviving in the 1980s before being marginalized again until 2012; Wiener's feedback

control was relegated to control engineering and not incorporated into AI's paradigm narrative; the probabilistic and information-theoretic traditions of the Soviet Kolmogorov school failed to enter the Western AI mainstream due to Cold War academic barriers and the paradigm exclusivity of symbolism.

These screened-out paths did not cease to evolve outside academia. This chapter focuses on a group of mathematical methods and researchers that found applications in finance—in quantitative investing, risk-adjusted out-of-sample performance is an important evaluative standard, with the Sharpe ratio as one representative measure, while pricing, risk management, and asset allocation have their own task-specific requirements: Monte Carlo entered derivatives pricing desks, Kalman filters entered statistical arbitrage, HMMs entered Renaissance's trading systems, convex optimization entered asset allocation—§8 provides a complete record of this migration.

The overall configuration of this period is **state selection + academic faction selection superimposed**: state selection determined which projects could obtain large-scale funding; academic faction selection determined which paradigms could enter the mainstream narrative. Capital channels were not yet the primary source of resources for AI during this period.

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§4 1990–2008: The Nineteen Years When the Primary Capital Selection Field Had Not Yet Done Work on the Research Subset Wearing the AI Label—The Control Baseline

These nineteen years are the control baseline period of this chapter. They are not nineteen years when "capital was absent"—quite the contrary, this was a period when both main lines—the computer industry (Wintel + Internet) and financial quantitative funds—simultaneously entered capital-driven high growth; but the research subset wearing the "AI" label (expert systems, connectionist basic research, symbolic reasoning theory) was barely within the primary selection field of Venture Capital (VC), while the subset of statistical learning and pattern recognition stripped of the "AI" label (search, e-commerce recommendation, information retrieval, quantitative trading, anti-fraud) was commercialized at scale under the primary driver of capital. Capital was not absent, but rather favored applications with identifiable commercial uses and visible paths to monetization, performing **selective absorption** of AI paths—the monetizable subset stripped of the "AI" label was absorbed; the non-monetizable subset was screened out. §8 will document a related but non-identical phenomenon: the scaled application in the financial field of mathematical traditions screened out by academia.

The Federal Funds Rate (FFR) underwent a complete cyclical fluctuation (the following are selected monthly observations, not annual averages or a complete series of extremes): 9.81% in May 1989 (a high observation of the preceding cycle, introduced as background) → 3.02% in 1993 (recession after the Gulf War and the Savings and Loan Crisis) → 6.50% in 2000 (peak of the dot-com bubble) → 1.00% in 2003 (post-dot-com crash bailout) → 5.25% in 2007 (peak of the housing bubble) → 0.16% at the end of 2008 (zeroed after the subprime crisis); monthly FFR data are from the Federal Reserve Economic Data (FRED) FEDFUNDS series of the Federal Reserve Bank of St. Louis (Federal Reserve Bank of St. Louis, 2026a), hereinafter the same; throughout this chapter, FFR refers to the monthly average effective federal funds rate in this series, and figures accompanying specific events are monthly averages for the month in which the event occurred—not event-day rates, nor the Federal Reserve's target rate. The observation cutoff for contemporary cases in this chapter is May 25, 2026; that year's financing negotiations, spending, and deployment plans are recorded as publicly known as of that date, not as full-year actuals.

The cyclical fluctuation of interest rates between 9.81% and near zero across two full rise-and-fall cycles constitutes the control period of this chapter's argument. However, the rise and fall rhythm of the research subset wearing the "AI" label from 1990–2008—the collapse of expert systems during the period when rates eased from 9% to 3%, the independent rise of statistical learning within academia in the mid-to-late 1990s, the emergence of the deep learning undercurrent in 2006 within the 4%–5% range—showed no visible coupling with the contemporaneous highs and lows of the FFR—these changes cannot be explained by contemporaneous interest-rate levels alone; the evidence examined here more directly identifies the roles of changes in academic beliefs and the computing power curve. The applied machine learning subset stripped of the AI label was absorbed at scale by the primary capital selection field during the same period, but its explosion cannot be explained by the tightness or looseness of interest rates alone—the evidence examined here more directly identifies the roles of internet data scale, electronic trading infrastructure, and monetizable business model opportunities.

The Continuation of the Second AI Winter (1989–1995). The commercialization of expert systems wearing the "AI" label collapsed at the end of the 1980s—the core problems were the knowledge acquisition bottleneck, runaway maintenance costs, and the failure of rules on the long tail. Lisp machine manufacturer Symbolics went bankrupt in 1993; Lisp Machines Inc. (LMI) went bankrupt in 1987; Thinking Machines Corp went bankrupt in 1994. These corporate failures extended from the late 1980s into the first half of the 1990s—the easing cycle of 1989–1993 covered some but not all of them: rates were easing, money was cheap, but expert systems stamped with the "AI" label had already been falsified by the belief market; cheap money could not save them. During the same period, capital did not leave the field: it turned to statistical pattern recognition not wearing the "AI" label—the Medallion Fund (Renaissance Technologies) was founded in 1988; Prediction Company (Farmer and Packard) was founded in 1991; PDT (Process Driven Trading under Morgan Stanley, formed by Peter Muller in 1992) entered a period of explosive profitability in the mid-to-late 1990s (Patterson, 2010, ch. 8). This is the key observation: **Capital selectively absorbed different methods and their applications; these cases do not isolate the effect of the AI label itself.**

The Rise of Statistical Learning and the Synchronous Absorption by the Capital Field (Mid-to-Late 1990s). During the same period, a path switch occurred within the AI academic community: from early projects in symbolism and connectionism toward statistical learning. Pearl's 1988 *Probabilistic Reasoning in Intelligent Systems* inaugurated the Bayesian network paradigm; Vapnik and Cortes proposed the Support Vector Machine (SVM) in 1995; Hidden Markov Models (HMMs) established dominance in speech recognition. An isomorphic absorption occurred within the capital field: in 1996–1997, the U.S. Securities and Exchange Commission (SEC) promulgated the Order Handling Rules, triggering the rise of Electronic Communication Networks (ECNs) such as Island and Archipelago, along with online brokers such as Datek; in 1998, Reg ATS (Alternative Trading Systems) established the compliance framework for electronic trading systems, and electronic trading gradually replaced floor-based market-making by the end of the 1990s; quantitative institutions such as D.E. Shaw (founded 1988), Two Sigma (founded 2001), and Citadel (founded 1990) hired large numbers of physics PhDs and mathematics PhDs during this period, deploying SVMs, Bayesian networks, HMMs, and genetic algorithms at scale (Bass, 1999; Patterson, 2010, 2012). Changes in both lines cannot be explained by interest rates alone; the more direct drivers were the maturation of statistical computing resources (PC + Internet + ECN) and the endogenous failure of the symbolic path.

The Last Glory of the Symbolic Path and the Dot-com Bubble Period. In 1997, IBM's Deep Blue defeated Kasparov—but this was a triumph of large-scale search plus an evaluation function, not of knowledge represen-

tation, and did not reverse the overall decline of the symbolic path that followed. The dot-com bubble peaked in 1998–2000, with the Nasdaq index rising from 1000 points in 1995 to 5048 points. The core beneficiaries of this capital feast were search, e-commerce, and portals—among them, Inktomi (distributed crawler and index), AltaVista (large-scale index and Boolean retrieval), Google PageRank (a stationary ranking computed from the web's link structure via normalized link transitions and a random-jump mechanism), and the collaborative filtering recommendation system that Amazon (founded in 1994) later built absorbed large amounts of VC funding. These methods would today be classified under information retrieval, statistical ranking, graph-theoretic ranking, and collaborative filtering—but were not named "AI" at the time. Cutting-edge AI research (symbolic systems, basic neural network research, probabilistic reasoning theory) was barely in the primary VC selection field during the first half of the 2000s—applied machine learning was within the VC field of vision but did not wear the "AI" label. Google, early on, used machine learning extensively as an internal tool, not as the story told to the outside world.

The Deep Learning Undercurrent and the First Quant Crisis (2006–2008). In 2006, Hinton, Osindero, and Teh published "A fast learning algorithm for deep belief nets," proposing the layer-wise pre-training method for Deep Belief Networks (DBNs), widely regarded as a key node in the revival of deep learning. In the same year, NVIDIA released the G80 architecture and the CUDA platform, inaugurating the era of general-purpose computing on Graphics Processing Units (GPUs). The FFR was then in the 4.29%–5.25% range—rates were neither particularly low nor particularly high, and no visible impact on the reception of Hinton's paper can be identified. Deep learning would not truly take off for another six years (2012 AlexNet). During the same period, the capital field experienced the first quant crisis: in August 2007, several quantitative funds based on factor models and statistical arbitrage suffered major drawdowns within days ("Quant Quake," Khandani & Lo, 2011); Regulation NMS (Reg NMS), adopted in 2005 and implemented in stages, was an important institutional component of the electronic transformation of U.S. equity markets; the 2008 global financial crisis followed immediately.

The Overall Configuration of These Nineteen Years: The evolution of AI paths was constituted by two simultaneous selections. Internal academic selection: symbolic path fails → statistical learning rises → deep learning undercurrent surges, driven by changes in academic belief and the computing power curve. Primary capital selection: applied machine learning stripped of the "AI" label (search, recommendation, quantitative trading, anti-fraud, information retrieval) was commercialized at scale; the expert system and basic research subset wearing the "AI" label was screened out—the driving force was the data scale and business model opportunities brought by the internet, with no apparent coupling to the contemporaneous highs and lows of the FFR. This is the control baseline for the subsequent argument: **the primary capital selection field beginning to act at scale on cutting-edge research paths wearing the AI label had to wait until after 2009, when ZIRP changed the capital environment, lowered the threshold for supporting long-horizon narratives, and the capital narrative of "the path to AGI" gradually became mainstream within the VC field around 2015.** During 1990–2008, capital supported AI-related methods mainly through specific commercial applications, while providing less support to some projects centered on AI research without clear commercial uses—the decoupling of the FFR cycle from the latter class of projects reflects differences in financing targets and commercial uses, not merely the presence or absence of an AI label. This selective absorption and the "alternative field migration" of §8 are two related but non-identical sub-mechanisms—the former describes the process by which capital screens AI subsets according to "can it be monetized"; the latter describes the process by which

mathematical traditions screened out by academia gain scaled application in the financial field; the two overlap in some periods but differ in driving forces and the planes on which they act.

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§5 2008–2022: A Prolonged Low-Rate Environment and the Expansion of Frontier AI Financing

After the 2008 subprime crisis, the Federal Reserve pushed interest rates to zero—holding them near zero for about seven years, from December 2008 until the first hike in December 2015; it then raised rates from December 2015 through December 2018, with the monthly effective rate peaking at 2.42% in April 2019 (FEDFUNDS); a cut was announced on July 31, 2019, and the monthly series reflected the change from August; in March 2020 it cut rates back to zero in response to the pandemic, holding them there until the hike in March 2022—about two years of near-zero rates—leaving policy rates overall at historically ultra-low levels. This prolonged low-rate period included two near-zero-rate episodes, separated by a phase of monetary-policy normalization. Chancellor (2022) re-emphasized an ancient insight obscured by the technicalization of modern finance: interest is the price of time. When the price of time approaches zero, the discounting gap between near-term and far-term spending narrows—advocates of zero rates describe this as capital becoming nearly infinitely patient. The precise mechanism is: zero interest rates lower the opportunity cost threshold for equity capital to support long-cycle projects—but patience is never truly infinite; its actual length is still jointly determined by the structure of funding sources (pension funds, sovereign wealth funds, corporate strategic investment) and exit market expectations.

Ma and Zimmermann (2023) estimate that VC falls by about 25% over one to three years following a 100-basis-point (bp) monetary tightening shock—the study measures the response period of aggregate financing after a tightening shock, supporting an effect of monetary conditions on innovation financing, but not establishing commercialization deadlines for different AI approaches. That VC showed greater tolerance for long-horizon narratives during the zero-rate period is consistent with this aggregate evidence in direction; this chapter does not state it as a sufficient condition for project financing.

The shape of the selection function during the Zero Interest Rate Policy (ZIRP) period was "the credibility of the future AGI narrative." Three representative samples can outline the contours of this shape: DeepMind and the early OpenAI show different forms of support obtained by long-horizon research organizations, while GPT-3 and its API show how earlier research investment entered commercial deployment during the low-rate period. DeepMind was founded in 2010 in an environment with FFR at 0.19%, and was acquired by Google in 2014 for approximately $500 million—that a pure research company with no revenue, continuously investing in basic reinforcement learning research (using Atari games as a testbed), could survive for four years and receive this valuation is a structural condition provided by zero interest rates. OpenAI, founded in 2015, had no revenue, no product, and was purely mission-driven. The GPT-3 paper was released in late May 2020, with the API following in June, in an environment with FFR between 0.05% and 0.08%.

The commonality between DeepMind and the early OpenAI is: they could clearly tell the story of "someday leading to AGI," but lacked the capacity to clearly state, within three years, "when and where, how many dollars will be monetized." The zero interest rate environment lowered the threshold for such narratives to obtain serious financing—because with the risk-free discount rate near zero, far-future uncertain returns remained substantial after discounting.

The ZIRP period was not one in which LLMs always dominated. After AlexNet in 2012, the connectionist resurgence suppressed symbolism—the main causes were GPUs, backpropagation, and large datasets; this change cannot be explained by contemporaneous interest-rate levels alone. In 2015–2018, reinforcement learning occupied the center (AlphaGo, AlphaZero, Dota, StarCraft), with DeepMind as the flagship. At that time, LLMs were not yet "the path": GPT-1 (2018) had received technical research attention, but scaling autoregressive language models had not yet become the dominant investment narrative toward AGI; GPT-2 (2019) was regarded by some commentators as a marketing gimmick.

Scaling, in the fields of physics and complex networks, is a broader framework of proportional relationships (as used in network topology by Barabási & Albert, 1999); building on prior empirical scaling studies such as Hestness et al. (2017), Kaplan et al. (2020) were the first to provide extrapolatable power-law fits and compute-budget allocation for neural language models spanning about seven orders of magnitude. Before that, "purely relying on increasing parameter count and data scale" was a path generally distrusted by LeCun, Marcus, the reinforcement learning camp, and the neuro-symbolic camp. After GPT-3 (mid-2020), the credibility of this path within the research community shifted markedly. That is to say: scaling/LLMs were a consistently doubted minority path before 2020. Their transition from a minority to the mainstream occurred between 2020–2022—the driving force was the empirical curve of GPT-3 markedly shifting the credibility of this path, superimposed on the tolerance for the compute bill provided by the zero interest rate environment.

The zero interest rate environment lowered the threshold for patient equity capital to support high-burn, far-horizon monetization paths, allowing multiple AGI-type organizations to coexist and experiment—rather than selecting which path is true. The capital environment's mode of action here is to allow multiple paths to survive and experiment simultaneously (altering the number of survivors)—not to select which path wins out; §7 will unfold the other mode of action: the shift of capital's financing criteria toward near-term commercial proof after 2022, altering the relative allocation among paths—the capital environment begins to directly select the type of path itself. The two modes of action require separate evidentiary support.

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§6 Boundary Conditions (Japan 1982–2025): Low Interest Rates Plus Balance Sheet Expansion Are Not Sufficient Conditions

§5 argued for the mechanism by which zero interest rates allowed multiple AGI paths to survive. But low interest rates plus significant balance sheet expansion alone are not sufficient conditions. Japan provides a control case spanning a complete long-cycle of interest rate decline.

The Bank of Japan's discount rate fell to 0.5% in September 1995 and to 0.10% in September 2001; the policy rate turned negative in January 2016, the negative rate ended in March 2024, and the policy rate rose to 0.25% in July 2024, to 0.5% in January 2025, and to 0.75% in December 2025; the central bank's total assets expanded from 80.07 trillion yen in April 1998 (the starting point of the FRED monthly series) to a peak of 764.81 trillion yen in August 2024, expanding approximately 9.5 times over 26 years (FRED JPNASSETS series, Federal Reserve Bank of St. Louis, 2026b). Japan experienced prolonged low interest rates and significant central bank balance sheet expansion, but this did not automatically replicate the organizational landscape of U.S. frontier AI financing. Japan in the same period also saw the emergence of foundation-model research organizations, but not a landscape of multiple technological paths competing in parallel with financing of comparable scale.

This section compares financing scale, organizational structure, and the capacity to sustain multiple lines of research—not whether such organizations exist at all; that low interest rates did not translate into multi-path frontier financing through Japan's capital structure reflects the problem of capital channels and industrial betting structures.

FGCS concentrated dedicated resources on parallel logical inference—a case of a national project betting on a specific technological architecture, but not representative of all Japanese computer R&D at the time. §3 already mentioned that MITI launched the Fifth Generation Computer Systems (FGCS) project in 1982, investing approximately 54 billion yen over more than a decade, betting on parallel logic programming. Sustained large-scale public investment did not guarantee broad commercial adoption; this chapter focuses on the roles of concentrated investment and technological-ecosystem competition, without ruling out other resource and execution constraints on the basis of budget totals alone. Its specialized architecture faced competition from general-purpose microprocessors, workstations, and established software ecosystems, and successor projects turned to porting part of the software to other platforms. The mechanism exposed here is: state capital's bets tend to be more concentrated—it inclines toward concentrating bets on a single path, with less room for error. The market at least allows multiple paths to each obtain seed funding—once national industrial policy picks a direction, the space for the remaining branches to acquire resources narrows substantially.

The financial system alienated "venture capital" into a tool for maintaining banking relationships. Kenney, Han, and Tanaka (2007) documented the institutional genetics of Japanese VC: between 1982–1984, city banks, securities companies, and regional banks together established 37 new venture capital subsidiaries. But "Japanese venture capitalists did not seek capital gains; rather they wanted to develop long-term banking relationships with their portfolio firms. Given these goals, due diligence did not have to be overly rigorous." In 1995, the government once again promoted reform. Thereafter, SoftBank, leveraging its U.S. internet profits, invested $8.8 billion in over 600 startups globally (among them hundreds of Japanese internet companies; by January 2001), but "since then, there has been little investment in start-ups." Kenney et al.'s conclusion is: the difficulties VC has had in taking root in Korea and especially Japan "seem to be intimately linked to the overall configuration of those societies and their political economies." Silicon Valley VC targets capital gains, tending to encourage diversified hedging-style bets; Japanese bank-affiliated VC targets relationship maintenance, tending toward perpetuating existing paths. These historical materials show that the allocation of low-cost funds was shaped by capital organization and established relational structures. They provide historical background for understanding the organizational conditions of Japan's later frontier AI financing, but by themselves do not establish that these mechanisms continued at the same strength through 2024. Kenney et al. describe the history of Japanese VC formation and its early transformation; they also record the emergence of independent VC and equity investment—how these early structures carried over into, or changed by, the frontier AI period requires separate treatment.

Low-interest funds were absorbed by zombie loans. Caballero, Hoshi, and Kashyap (2008) provided the measurement: banks used low-cost funds to pseudo-restructure borrowers that should have gone bankrupt—"evergreening"—continuing to feed subsidized credit to zombie firms. In the early 2000s, in the major sectors examined (manufacturing, construction, real estate, retail, wholesale, and services), approximately 30% of listed companies, and approximately 15% of total assets, were in a zombie state. The consequence was not resource scarcity, but that zombies depressed market prices and raised labor prices, making it harder for healthy firms to enter and invest—"even solvent banks saw no particularly good lending opportunities in Japan." Kobayashi and

Inaba (2002) examined, at the general equilibrium level, another transmission path of non-performing loans: the "complexity externality"—coordination failures in the division-of-labor network; the inefficiency of one firm diffuses through the supply chain to the productivity of other firms, and the economy as a whole enters a bad equilibrium. This mechanism is explicitly not a shortage of capital supply, nor soft budget constraints, but a coordination failure within the division-of-labor network.

The two layers superimposed give the boundary condition for the proposition of §5: **in terms of the U.S. AI capitalization experience of 2009–2021 discussed in this chapter, low interest rates are an important condition but clearly not a sufficient condition.** Whether monetary conditions can be converted into effective selection through the capital environment depends on at least three preconditions. First, the target path must enter the primary capital selection field—for the frontier research examined in this chapter, this meant wearing the "AI" narrative label in the period concerned; second, the capital channel must permit independent technical judgment, risk-taking, and sustained research support—the issue is governance and investment orientation, not whether an investor is bank-affiliated; third, the industrial organization and capital channels must not be locked in by path dependencies formed before the low-interest-rate period—Japan responded to IT and cutting-edge AI industries with the catch-up model of traditional cyclical industries (state concentrated bets on a single technological path + bank-affiliated VC targeting relationship maintenance rather than diversified bets), and the catch-up model failed to achieve sustained success in the IT industry; this path dependency had already compressed the space for multi-path parallelism before the 1995 low-interest-rate period arrived. The three conditions are the boundary conditions this chapter focuses on. Monetary conditions are not a single-variable referee, but one of the upstream constraints coupled with the capital channel structure, industrial policy, and narrative credibility.

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§7 2022–Present: Financing Divergence after Monetary Tightening

Starting in 2022, the Federal Reserve, to suppress inflation, initiated the steepest rate hikes in forty years: the FFR rose from 0.08% (February 2022) to 5.33% (pegged from August 2023 to August 2024). This is not a continuation of ZIRP—it is a switch of the selection function.

The high interest rate environment does not stop selection—it changes the type of path capital is willing to support. The selection function during the ZIRP period was "the credibility of the future AGI narrative"; under the high interest rate period it has shifted toward near-term commercial validation. Monetary tightening increased some external investors' emphasis on near-term user adoption, willingness to pay, and cost structure, without imposing a uniform commercialization deadline on all AI projects.

The victory of LLMs is not because monetary conditions had no effect, but because they precisely met capital's new financing criteria. When ChatGPT launched in November 2022, the FFR had already risen to 3.78%. According to contemporary estimates from research institutions, it reached 100 million monthly active users within about two months of launch, demonstrating large-scale user demand. OpenAI received a new multi-year investment from Microsoft in January 2023 (widely reported by media at a total of approximately $10 billion), with FFR = 4.33%. ChatGPT's early rapid growth and Microsoft's expanded investment occurred in succession, showing the compatibility between user adoption and strategic investment; publicly available materials are insufficient to separately confirm the full basis for this round of investment decisions.

The symmetry of the counterexample: paths that cannot be commercialized under the current capital environment face financing difficulties. Under the high interest rate environment, projects that can only tell the story of "someday leading to AGI" without a clearly articulable near-term commercialization path find it harder to renew capital at the same scale. Some pure research organizations founded during the ZIRP period saw financing-model adjustments between 2023–2024, with some pivoting to the LLM application layer or being acquired; "pure research" organizations like DeepMind were asked within the parent company Google to merge with Google Brain (forming Google DeepMind in April 2023) and take on more direct productization tasks; Anthropic's valuation is high, but its financing cadence is tightly bound to its commercialization progress (Claude enterprise API revenue).

Embodied intelligence exploded during the high-rate period precisely because it (in part) offered a commercialization story that could be told clearly. The data from 2024–2026 are compatible with investors' emphasis on application direction, while also showing that exit windows, industrial strategy, and long-term technological expectations were still at work. Figure AI was valued at $39 billion, with a Series C exceeding $1 billion (Figure AI, 2025); its publicly announced large-scale deployment plans and industry investors constitute early market signals; China's embodied intelligence sector raised 39.8 billion yuan in 2025, a year-on-year increase of 326% (OFweek, 2026).

The key prior accumulation was mostly completed during the ZIRP period—Transformer (Vaswani et al., 2017), Diffusion, Sim2Real took shape during the low-rate period; the naming and representative public models of Vision-Language-Action (VLA) crossed into the high-rate period—but the companies were left standing under the high interest rate environment by the screen of "can tell a clear commercialization story."

The Exceptional Channel: Reputational Capital Exceptional Financing. AMI Labs, founded by LeCun, raised $1.03 billion in March 2026, with a pre-money valuation of $3.5 billion and a post-money valuation of approximately $4.5 billion (TechCrunch, 2026a; MIT Technology Review, 2026). However, AMI does not belong to the "commercial proof" model: its CEO LeBrun publicly stated that the company starts with fundamental research and that it could take years to go from theory to commercial applications, explicitly rejecting the applied AI path of "release a product in three months, have revenue in six months." This is an exceptional channel for capital selection under the high interest rate environment—Turing Award-level scientific reputational capital can substitute for commercial proof, allowing an explicitly long-term pure research project to receive funding of an entirely different order of magnitude for this round under an environment with FFR at 3.64%.

The Boundaries of the Exceptional Channel. The observed cases of this channel display three noteworthy covarying characteristics: (i) scientific reputation—which helps secure research judgment space independent of commercial metrics; (ii) a public commitment to long-term basic research not organized around near-term productization—so that the project is not evaluated as applied AI within the selection function; (iii) substantial initial funding—which may extend the financing buffer. These are relevant characteristics observed in these cases, not established necessary conditions. The confirmed sample of the same class observed by this chapter is LeCun's AMI Labs; among other projects described by media as "reputational exemption" (such as Sutskever's SSI), SSI announced large-scale financing in 2024 and an explicit long-term research orientation; its public profile fits the characteristics of this channel. Physical Intelligence in the embodied intelligence direction shows a similar configuration: it had a completed round valuation of approximately $5.6 billion in 2025, and media reported in 2026 that it was negotiating new financing at a valuation exceeding $11 billion (Bloomberg, 2025; TechCrunch, 2026b), while its co-founder publicly stated that the company has no timeline

for commercialization—the financing flows to long-term research rather than near-term monetization, supported by investors' conviction in the foundation-model route rather than verifiable commercial proof. In their early stages, OpenAI and Anthropic were closer to the "grand narrative" screening period named in this chapter than to the "reputational exemption" channel of the high-rate period—the two period-specific financing configurations are this chapter's naming distinction, not mutually exclusive causal claims. Beyond external financing, internal budgets or support from affiliated organizations may provide long-horizon research with time conditions different from those faced by conventional VC-backed projects—the organizational link between High-Flyer and DeepSeek is a case in point (see §9).

Taken together, this chapter has focused on two external financing logics under the current capital environment: (a) those that have already delivered commercial proof (the OpenAI/ChatGPT channel); (b) pure research projects supported by extremely high scientific reputational capital and an explicit long-term orientation (the LeCun/AMI channel)—substantial initial funding can provide a sizable financing buffer. The remaining paths find it harder to obtain capital support of the same scale and patience under the current financing environment.

The Expert-System Commercial Crisis as a Contrast. §4 has already shown: the collapse of expert systems was an internal falsification of technical belief that easy money could not save—when the commercialization capability itself does not stand, the length of the interest rate window is not a decisive variable. This precisely serves as a contrast to the situation after 2022: whether AI can pass the capital screen still depends on whether the commercialization capability itself can stand, only now capital's financing criteria have concentrated on near-term commercial validation under the high interest rate environment.

What Is Locked Is Infrastructure, Not Epistemological Truth. What capital selected under the high interest rate environment are "paths that can be commercialized"; what is locked is the infrastructure corresponding to this path: four tech giants are planning expenditures of approximately $650 billion in 2026 (Bloomberg, 2026), and Stanford HAI (2026) recorded the expansion of AI corporate investment—reaching $581.7 billion globally in 2025, with generative AI accounting for nearly half of private AI funding (the tally includes financing and M&A, and is not equivalent to new infrastructure spending)—record investment completed under an environment where the FFR had fallen back from its 2023–2024 highs (no higher than 4.33% throughout 2025). This investment reinforces infrastructure for large-scale neural-network computation; it is not exclusive to language models—embodied intelligence likewise relies on large-scale matrix operations—but it may favor approaches that can readily use the existing hardware and software stack.

But what is locked is infrastructure, not the truth structure of intelligence. What monetary conditions change through the capital environment is "which path first delivers commercial proof, which path gets built into infrastructure"—not "which path naturally wins out epistemologically."

Early Signals of a Return of National Security Selection in 2026. On January 9, 2026, the U.S. Department of Defense—which since September 2025 has used War Department (Department of War) as an authorized secondary designation by executive order—released the "Department of War Artificial Intelligence Strategy" memorandum, explicitly positioning the Department's goal as building an "AI-first" warfighting force through seven "Pace-Setting Projects" (U.S. Department of War, 2026a). On May 1 of the same year, the Department of War announced classified network (IL6/IL7) AI deployment agreements with eight cutting-edge AI companies (SpaceX, OpenAI, Google, NVIDIA, Reflection, Microsoft, Amazon Web Services, Oracle) (U.S. Department of War, 2026b)—marking a significant upgrade and publicization of national security selection in 2026: An-

thropic had stated in February 2026 that its models were deployed on U.S. government classified networks (Anthropic, 2026), and on December 22, 2025, the Department announced an agreement with xAI, targeting initial IL5 deployment on GenAI.mil in early 2026 (U.S. Department of War, 2025). Notably, according to Anthropic's statement, the disagreement concerned two categories of use restrictions—mass domestic surveillance of Americans and fully autonomous weapons; Anthropic insisted on retaining the relevant use guardrails, the negotiations broke down, and it was not selected—the other side of the selection function is thus visible: it screens not only "whose technology is strong enough" but also "who accepts 'any lawful use' contract language." This development may signal the return of national security selection in a new form—no longer the ARPA-style practice of §2, funding people rather than projects, but the government directly acting as a classified-level client for cutting-edge AI companies. If this trend continues, the relationships among the four selection mechanisms of this chapter may undergo a new round of switching: capital selection and national security selection may return from the current substitution relationship to a superposition relationship. This paragraph serves only as a signal record and does not yet constitute a full argument.

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§8 Orthogonal Selection of the Alternative Field: The Eighty-Plus-Year Migration History of Silenced Paths

§3 has already shown how paths such as statistical pattern recognition, connectionism, and cybernetics lost influence, were diverted, or were renamed within the mainstream AI narrative in different phases—the periods of marginalization differed across traditions. A deeper question is: where did these paths that lost influence in the mainstream go?

The Orthogonality of the Two Sets of Selection Functions. The selection criterion of academic AI: "does this look like intelligence?" Statistical pattern recognition "doesn't look like intelligence," so it was eliminated; von Neumann's self-reproduction "doesn't look like intelligence," so it was relegated to automata theory; Wiener's feedback control "doesn't look like intelligence," so it was relegated to control engineering.

In quantitative investment, risk-adjusted out-of-sample performance is an important standard—it does not care whether the model "understands" anything, only whether it can keep making money; pricing, risk management, and asset allocation each have their own task requirements—finance does not have a single selection function covering all tasks.

The result: a batch of methods that lost influence in the mainstream AI narrative gained sustained organized application in finance, an orthogonal selection field.

Von Neumann Path → Complexity Economics → Quantitative Hedge Funds. Alongside his work on stored-program computing, von Neumann in his later years (1948–1957) also delved deeply into self-reproduction, evolution, cellular automata, and complexity. His posthumous manuscript *Theory of Self-Reproducing Automata* (1966, edited and published by Arthur Burks) pointed toward a research program: the essence of computation concerns complexity and self-organization, not symbol manipulation. This path was marginalized in the AI academic community for fifty years.

In 1987, the Santa Fe Institute (SFI) held a conference: Nobel laureate in economics Kenneth Arrow brought theoretical economists; physicist Philip Anderson brought complex systems scientists. The conference's out-

come was SFI's first formal research program, "The Economy as an Evolving Complex System"—led by W. Brian Arthur (Anderson, Arrow & Pines, 1988; Arthur, 2021).

This complex systems path has clear connections to the talent and method networks of several subsequent quantitative institutions. J. Doyne Farmer and Norman Packard—members of the "Dynamical Systems Collective" at UC Santa Cruz in the 1970s, co-authors of the foundational chaos theory paper "Geometry from a Time Series" (Packard et al., 1980)—left their positions respectively as head of the Complex Systems Group at Los Alamos National Laboratory (Farmer) and tenured professor and member of the Center for Complex Systems Research at the University of Illinois at Urbana-Champaign (Packard) to found Prediction Company in Santa Fe in 1991, one of the earliest companies to perform fully automated quantitative trading (Bass, 1999, ch. 1–3). The prior careers of both men were in chaos theory, artificial life, and complex systems—both belong to the subsequent work within the complexity and self-organization problem domain opened up by von Neumann; the inheritance relation between their specific methods and that program must still be judged on the basis of citations and research records. Prediction Company signed an exclusive research contract with Chicago derivatives market-maker O'Connor & Associates; in 1992, O'Connor was acquired by Swiss Bank Corporation (SBC), and the contract was transferred into SBC; in 1998, SBC merged with Union Bank of Switzerland to form UBS, into which the contract was subsumed; it was formally acquired by UBS in 2005 (Finextra, 2005) and transferred to Millennium Partners in 2013. Farmer later developed the theory of "market ecology" at Oxford University: market participants are viewed as species in ecological niches, with predator-prey relationships between trading strategies—this is precisely the unfolding in finance of von Neumann's "self-organizing system" program.

Another direct descendant of the von Neumann path is John Holland—a student of Burks, who systematically expounded genetic algorithms in his 1975 monograph. Holland joined SFI's economics program; genetic algorithms were extensively used by firms such as D.E. Shaw and Prediction Company in the 1990s for parameter space searches of trading strategies.

Ulam Monte Carlo → Derivatives Pricing. Stanislaw Ulam was von Neumann's collaborator at Los Alamos and the co-inventor of cellular automata. The Monte Carlo method he invented in 1946 for nuclear weapons simulations is today widely used as a foundational method in derivatives pricing on Wall Street. Phelim Boyle's 1977 paper in the *Journal of Financial Economics* pioneered the application of Monte Carlo to option valuation—a byproduct of Cold War nuclear simulation from thirty years earlier became the pricing engine of global financial markets.

Wiener Cybernetics → Black-Scholes-Merton and the Kalman Filter. Wiener's cybernetics was marginalized within the AI academic community by the symbolists after the 1956 Dartmouth workshop. Yet its core technologies—feedback control, stochastic processes, state estimation—entered the core toolkit of modern financial mathematics. Merton's (1969) continuous-time portfolio theory is essentially Bellman dynamic programming (a product of the Cold War RAND Corporation) transported into finance. The dynamic hedging of the Black-Scholes-Merton option pricing model (1973) belongs, in mathematical structure, to the class of continuous-time stochastic control pioneered by Wiener. The Kalman filter—proposed by Kalman in 1960 and subsequently applied by NASA to Apollo navigation—is used today by many statistical arbitrage strategies.

Cold War Cryptography → Renaissance Technologies. Renaissance Technologies was founded by James Simons in 1982. Simons left the Harvard mathematics department in 1964 to join the Institute for Defense Analyses (IDA) in Princeton, where his work involved assisting the National Security Agency (NSA) in monitoring

and decrypting Soviet military ciphers (Zuckerman, 2019, ch. 2); his IDA colleague Leonard Baum, during the same period, co-invented the Baum-Welch algorithm with Lloyd Welch—the core method for training Hidden Markov Models (HMMs). In the autumn of 1967, Simons wrote a letter to the *New York Times* opposing the Vietnam War, and was dismissed by IDA after subsequently speaking about his work arrangements in a media interview (ibid., ch. 2).

Renaissance recruited mathematicians, physicists, cryptographers, and astronomers, barely overlapping with Wall Street's traditional recruitment pool; its methodology was to detect patterns in noise, sharing the same origin as signals intelligence practice. The Baum-Welch algorithm, invented during IDA cipher-breaking work, later became one of the landmark methods of twentieth-century machine learning; Simons and Baum brought the hidden Markov methods of the IDA period into their early trading practice in the late 1970s (Zuckerman, 2019). These mathematical tools and their personnel, trained during the Cold War, had already gained organized application in the financial field in the 1970s–1980s, before the Cold War ended—what can be confirmed is the personnel link between cryptographic research and early quantitative trading, not a causal role for the Cold War's end.

The causal structure of this migration chain directly echoes §§2–3: the Cold War created institutions like IDA → internal Cold War contradictions (Vietnam War politics) screened out Simons → Simons entered finance carrying the mathematical tools trained by the Cold War → Renaissance became the most successful quantitative fund of the late twentieth century. **A byproduct of national-level selection flowed into the financial field.**

Kantorovich and Markowitz: The Same Mathematical Family in Different Institutional Settings. Soviet mathematician Leonid Kantorovich invented linear programming in 1939, eight years before Dantzig. The full-scale promotion of this work long faced ideological resistance in the Soviet Union—its "resolving multipliers" (i.e., shadow prices) were seen by orthodox Marxist economists as a dangerous concession to market price theory. Kantorovich himself was a member of the Soviet Academy of Sciences, a recipient of the Lenin Prize and the Stalin Prize, yet when he won the Nobel Prize in Economics in 1975, the prize was awarded precisely for this optimization theory that had been difficult to fully promote within the Soviet Union. Markowitz's mean-variance portfolio theory of 1952—today the mathematical basis for everything from BlackRock's Aladdin to global pension asset allocation—is essentially a quadratic programming problem, belonging to the same convex optimization family as the linear programming developed by Kantorovich-Dantzig in the 1940s (in his Nobel lecture, Markowitz traced his intellectual sources to Williams's 1938 *Theory of Investment Value* and von Neumann-Morgenstern's 1944 expected utility theory; this chapter does not claim, on the basis of mathematical family resemblance, a demonstrated direct line of transmission from Kantorovich to Markowitz). These cases show applications of the same mathematical family in different institutional settings—convex optimization hindered by ideology in the Soviet Union, and a standard tool in U.S. asset allocation; this chapter does not equate the family connection with a demonstrated direct line of transmission.

The Adoption of Statistical Learning, Neural Networks, and Evolutionary Search in Quantitative Trading. Neural networks were marginalized in the AI academic community from 1969 onward (Minsky and Papert's 1969 *Perceptrons* was the direct trigger; the tradition briefly revived in the 1980s before being marginalized again until 2012). Yet in the financial domain, statistical pattern recognition was adopted at scale from the late 1980s onward—finance does not care whether a model has "understanding" or "reasoning," only about out-of-sample performance. Simons and the IDA code-breakers had proposed using hidden Markov models to predict stock prices in a paper as early as the late 1960s (Zuckerman, 2019, ch. 2), and his early quantitative practice

(Axcom, mid-1980s) further modeled the market as a Markov-like process (ibid., ch. 4); in 1993, Renaissance recruited Peter Brown and Robert Mercer from IBM's speech-recognition and statistical machine translation team to lead the machine learning direction (ibid., ch. 9); the two later became co-CEOs of Renaissance; D.E. Shaw, founded in 1988 by Columbia University computer science professor David Shaw, used neural networks and genetic algorithms in the 1990s (Bass, 1999, ch. 14); Prediction Company, from 1991 onward, used neural networks, genetic algorithms, and other methods to perform fully automated trading (ibid., ch. 7). Two Sigma (founded in 2001 by former D.E. Shaw members John Overdeck and David Siegel) and Citadel Securities (which spun off its market-making business in 2002 from the Citadel hedge fund founded in 1990), as subsequent continuators of this methodological tradition, further scaled similar methods.

The timing of financial institutions' scaled adoption of statistical learning methods partially overlapped with these methods' development in academia—not because finance is more forward-looking than academia, but because finance's selection function never passes the gate of "does this look like intelligence," only looking at out-of-sample performance; the difference lies in evaluative objectives and application conditions, and cannot be summarized as finance accepting deep learning twenty years before academia.

These cases suggest a common direction: mathematical traditions that lost influence in one evaluative setting could gain organized application in another with different criteria—the cases differ in evidential type and strength, and should not all be treated as a single direct transmission chain.

Research Tradition and Cross-Field Connection	Scaled Adoption in Finance	Core Mathematics
Von Neumann Complex Systems / Evolution	SFI Complexity Economics / Prediction Company	Adaptive systems, genetic algorithms
Ulam Monte Carlo	Standard method for derivatives pricing	Path simulation
Wiener Cybernetics / Feedback	Black-Scholes-Merton / Kalman filter	Continuous-time stochastic control
Cold War Cryptographic Research (U.S. IDA)	Renaissance / Statistical arbitrage	HMM
Kantorovich Linear Programming	Markowitz / Asset allocation	Convex optimization
Holland Genetic Algorithms	Strategy parameter optimization	Evolutionary search
Statistical learning, neural networks, and evolutionary search	Renaissance / D.E. Shaw / Prediction Company quantitative trading	Pattern recognition, neural networks, evolutionary search

A conclusion emerges: **being screened out by one field does not mean the disappearance of that approach—an alternative field with markedly different evaluative criteria can provide new room for survival.** Academic AI screens for "the feel of intelligence"; finance evaluates risk-adjusted out-of-sample performance—the two sets of criteria differ, and paths eliminated by the former may gain organized application in the latter; the cases differ in evidential type and strength and should not all be treated as a single direct transmission chain.

This finding has symmetrical significance for the lock-in judgment of §7: the current LLM path is locked in relative to the selection field of "high interest rates + commercial viability + LLM infrastructure." Once another field with a selection function orthogonal to this one emerges—for example, the next-generation physical

engineering field, the national security field, or a field not yet visible—paths currently locked into secondary positions could equally find scaled application. **Lock-in is relative to a specific selection field, not an ontological termination.**

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§9 Mathematical Connections Across Fields and Their Limits

Financial engineering and deep learning share some mathematical tools and forms. The table distinguishes shared frameworks, similarities in recursive form, and functional analogies; historical transmission requires separate evidence concerning people, organizations, or documents.

Finance	AI	Shared Mathematical Structure	Relationship Grade
Geometric Brownian Motion $dS = \mu S\,dt + \sigma S\,dW$	Stochastic Gradient Descent (SGD) stochastic modified equation $d\theta = -\nabla L\,dt + \sqrt{\eta}\,\Sigma^{1/2} dW$	Both can be placed in the Itô diffusion framework, but with different drift and diffusion terms	Shared mathematical framework (the weak-order approximation holds between discrete SGD and its stochastic modified equation, see Li, Tai & E, 2017; not an approximation relation between the two)
Markowitz Mean-Variance Optimization $w^* = \arg\min_{\mathbf{1}^T w=1} w^T \Sigma w$	Attention weights $\alpha = \text{softmax}(QK^T/\sqrt{d_k})$	Rhetoric of weighted aggregation (mathematical objects differ: the former is the global minimum-variance portfolio under the budget constraint (with a closed-form solution under conditions such as a positive-definite covariance matrix and unrestricted short selling); the latter is a deterministic computation of softmax, with objective functions and constraints that differ in meaning)	Rhetorical similarity

Finance	AI	Shared Mathematical Structure	Relationship Grade
GARCH Volatility $\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$	Adam Optimizer $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$	Second-moment exponential weighting	Similarity of recursive form (both are exponentially weighted updates of squared terms)
Arbitrage Pricing $E(r_i) = r_f + \sum \beta_{ik}\lambda_k$	Recommender System Matrix Factorization $\hat{r}_{ui} = \mu + b_u + b_i + q_i^T p_u$	Rhetoric of low-dimensional factor decomposition (in APT practice, observable factors are commonly used to estimate β; matrix factorization factors are latent variables)	Functional analogy

Li, Tai, and E (2017) proved: in the limit as the learning rate $\eta \to 0$, there exists a weak-order $O(\eta)$ approximation between discrete SGD and its corresponding Stochastic Differential Equation (SDE)—the error in the expectation of test functions is bounded, but it is not a pathwise strict equivalence; here Σ denotes the covariance of per-sample gradients relative to the full-sample average gradient (the gradient noise covariance) (Li et al., 2017, Eq. 4). Mandt, Hoffman, and Blei (2017) further showed that, under conditions such as a locally quadratic objective and approximately stationary gradient noise, the stationary distribution of SGD with a constant learning rate can be interpreted as an approximation to a Bayesian posterior. This is not a metaphor—it is a provable mathematical approximation, at a grade higher than analogy but lower than exact equality of both sides. The weak approximation above concerns discrete SGD and its stochastic modified equation. The financial model and the SGD approximation in the first row merely share the Itô diffusion framework; no approximation relation between those two models has been established here.

Financial engineering and deep learning shared the same pool of physics PhDs from the 1990s through the 2010s: after completing their PhDs, some entered Wall Street to do quantitative trading; some entered machine learning research. The same set of mathematical traditions screened out of the AI mainstream entered finance through their career choices and re-entered AI through cross-field backflow—the quantitative private fund High-Flyer formally commissioned its large-scale proprietary computing cluster "Firefly One" in 2020 and continued to expand its computing infrastructure thereafter (the Firefly clusters; High-Flyer, n.d.); DeepSeek, founded in 2023, describes itself as a company dedicated to making AGI a reality (DeepSeek, n.d.). High-Flyer and DeepSeek were founded successively by the same person (Associated Press, 2025); this organizational link provides a contemporary instance of the backflow: computing and engineering resources accumulated in finance moved into foundation-model research. This case evidences the cross-field flow of contemporary organizational resources, not the direct transfer of any particular quantitative method, and it cannot by itself explain the earlier-formed mathematical structural relationships in the table.

Sharing similar mathematical tools often also brings similar blind-spot structures. Mandelbrot and Hudson's (2004) analysis of Dow Jones Index data from 1916–2003: for days with single-day moves exceeding 3.4 percent, the normal distribution assumption expected 58 days; 1001 such days actually occurred. Some parameter-

ized risk models (such as the variance-covariance approach to Value at Risk (VaR)) and Black-Scholes pricing adopt normal (log-normal) distribution assumptions—when the tails of the actual return distribution differ substantially from these particular specifications, they may underestimate the probability of extreme movements. This phenomenon suggests that model specifications may limit the scope of judgment—the training and inference of LLMs likewise depend on specific distributional assumptions, and their failure modes warrant dedicated examination; this chapter treats the two only as an analogy of model limitations, not as a mathematical isomorphism. The shared use of mathematical modeling tools prompts separate examination of each model's domain of validity, rather than a presumption that their failure mechanisms are identical.

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§10 Conclusion: The Overall Configuration of Eighty-Plus Years

Viewing the selection mechanisms of the five periods side by side, the eighty-plus-year history can be compressed into one sentence: **what has shaped the divergence of AI paths and the allocation of resources is not a single variable, but the ebb and flow of four selection mechanisms across different periods, organized by three relationships: superposition, substitution, and orthogonality.**

- **1943–1969**: The period of state-funding dominance. The selection function was "does it serve national strategy." Interest rates were not in the selection function.
- **1969–1989**: State selection and academic faction selection superimposed. The academic selection function was "does this look like intelligence as defined by this paradigm"—which long kept statistical pattern recognition, connectionism, cybernetics, and the Soviet probabilistic tradition at the margins of the mainstream AI narrative. Symbolic AI entered its golden age in a high-rate era with FFR between 6%–19%, with no stable correspondence to the level of interest rates themselves.
- **1990–2008**: The primary capital selection field performed selective absorption of AI. The statistical learning/pattern recognition subset stripped of the AI label (search, recommendation, quantitative trading, anti-fraud), together with financial quantitative institutions (Renaissance, Prediction Company, D.E. Shaw, Two Sigma, Citadel), became the objects of large-scale absorption by the capital field; the two violent cyclical fluctuations of the FFR, from highs around 9.8 percent to near zero, were uncoupled from the trajectory of the research subset wearing the AI label. The collapse of expert systems, the sinking of connectionism, and the undercurrent of deep learning were primarily driven by internal academic pressures and the computing power curve; the migration of mathematical paths into the financial field documented in §8 is a related but non-identical sub-mechanism.
- **2008–2022**: A prolonged low-rate environment allowed the capital selection field to absorb cutting-edge AI at scale for the first time. Financing criteria were more permissive of long-horizon, high-burn, far-future-monetization AGI-type narratives; the organizational forms and research paths in which DeepMind, OpenAI, and GPT-3 were situated could more readily obtain sustained support in this environment. However, the control case of Japan 1982–2025 defines the boundary: low interest rates plus balance sheet expansion alone are not sufficient conditions—it also requires capital channels that allow multiple paths to run in parallel and not be overridden by a stronger state selection or path-dependency structure.
- **2022–Present**: The high interest rate environment made capital place greater weight on near-term commercial validation. LLMs, aided by ChatGPT, delivered product-level adoption and commercial proof earlier; some embodied intelligence projects still received large-scale financing during the high-rate pe-

riod on the strength of deployment intentions and industry-investor endorsement; pure research paths find it harder to renew capital at the same scale under the current capital environment; some projects with scientific reputations, long-term funding commitments, or internal resource support continued to obtain substantial financing (AMI Labs and SSI as comparable cases supported by public evidence). National security selection also shows early signals of a return (see the end of §7 for details); capital selection and national security selection may re-enter a superposition relationship from the current substitution relationship (signal record only, no magnitude predicted).

Among the four mechanisms, only one—capital selection—has its effectiveness premised on "the target path entering the primary capital selection field." The degree to which this premise was satisfied varied across different periods: between 1943–1989, the capital field's involvement in computing and AI research was limited; between 1990–2008, the capital field entered via selective absorption, but the research subset wearing the AI label was still not on the main plane of capital's action; only after 2009 did the capital selection field begin to act at scale on "paths wearing the AI label" themselves. "Monetary conditions alone do not explain the rise or decline of any category of AI paths"—during 1990–2008, the applied machine learning subset stripped of the label did achieve large-scale commercialization amid capital expansion, but its growth depended equally on internet data scale, electronic trading infrastructure, and monetizable business models; monetary conditions were only one upstream layer of that background.

The second overall configuration comes from the orthogonal relationship: **paths screened out by one field can gain scaled application in another field whose selection function is orthogonal to the original field.** Von Neumann complex systems, Ulam Monte Carlo, Wiener cybernetics, Kantorovich convex optimization, Holland genetic algorithms, Cold War cryptographic mathematics—a number of these methodological traditions that lost influence in the mainstream AI narrative gained more stable, and also higher cash-flow, organized application in the financial field. This is not proof of a single chain, but a set of historical samples differing in evidential type and strength, yet consistent in direction.

This finding has symmetrical significance for the current LLM lock-in: **lock-in is usually relative to a specific selection field, not an ontological termination.** Tang Shiping (Tang, 2013, ch. 1) has already given an independent methodological diagnosis of this point—"inheritance of ideas in human society often, if not always, requires the backing of power"—he left room for the non-universal quantifier with the qualifier "often, if not always"; this chapter accepts this qualification: the current LLM path, locked in relative to the selection field of "high interest rates + commercial viability + LLM infrastructure," is a position endorsed by power (capital power + the commercial viability selection function) under the current selection field, not an epistemological termination of truth. Once another field with a selection function orthogonal to this one emerges—the next-generation physical engineering field, the national security field, or a field not yet visible—paths currently locked into secondary positions could equally regain scaled application. Silenced paths, changing names, changing selection fields, can reappear. An alternative field with markedly different evaluative criteria can provide new room for survival for disfavored paths—this chapter calls such a relationship "orthogonal" (by analogy), but does not treat it as a necessary condition for all cross-field survival.

The LLM is the current winner of over eighty years of path selection. The large-scale deployment and continued expansion of LLMs depend on computing infrastructure, training data, and market adoption, among other conditions; interest-rate cycles affect the financing environment for these activities; they do not supply their technical principles. Its ability to fluently recount the stories of silenced paths itself carries a structural bias: the

"silenced" it can see is the part the current narrative permits it to see. Paths truly silenced to the point of leaving no written records—are as invisible to it as to the reader. This invisibility is not a posture of the chapter's author; Tang Shiping (Tang, 2013, ch. 1) has already given an independent methodological diagnosis of this point—"movements in the other direction has been invisible to us (i.e., many social entities have disappeared and left few traces behind)." This is the farthest this chapter can reach. To go further requires not more search, but the acknowledgment that a boundary of structural invisibility has been reached.

Appendix: A Note on Claims of the Type "AI Hardware Improves X Times Per Year"

(Note: a calibration analysis only, not constituting part of the main argument.)

Under the high interest rate environment, capital takes "commercial viability" as its selection criterion; whether this criterion changes how upstream hardware suppliers characterize their products' progress is a question on which this chapter takes no prior stance—the following is a calibration analysis of one industry-basis sample from 2026:

He (2026) states the time-scaling factor for AI workloads as "up to 10× per year, where throughput translates directly into economic value." Read strictly according to its original qualifiers: in the paper's cross-layer generational rule $\tau_{n+1} = \tau_n/\alpha$, the scaling factor α is application-specific rather than universal; "up to 10×" is the upper-bound value for AI workloads, drawn from production experience, and the clause indicates the correspondence between throughput and economic value in AI settings—the driving background of that value, not the definition of the parameter itself.

On the other hand, independent engineering benchmark measurements are entirely different: Hennessy and Patterson (2019), based on SPECintCPU, give "End of the Line ⇒ 2X/20 years" (annotated in the original as 3%/yr); in their historical review of general-purpose processor integer benchmarks, they characterized the final phase of performance growth as approximately 3% per year; this benchmark does not represent all hardware classes or directly report growth through 2026—the two figures differ in application scope, workload, and measurement basis (the former is a production-experience upper bound on time-constant scaling in a specific application domain; the latter is a historical statistic of general-purpose processor integer benchmarks), and no difference in multiples of direct engineering significance can be calculated from them.

These two sets of figures do not contradict each other; they answer different questions: the former is "by what factor at most does the time constant shrink each year for a specific application (AI workloads)" (production-experience upper bound, industry basis); the latter describes the annual growth trend of a specified processor-performance benchmark in the historical series examined (independent engineering basis)—reading the former as "AI hardware improves 10 times per year" would constitute a misrepresentation. The purpose of this analysis is not to endorse such industry-basis claims but to remind readers that the two figures cannot be read across each other; the internal logic of this chapter's argument on selection mechanisms does not depend on any single piece of industry self-reporting text as external corroboration.

References

Anderson, P. W., Arrow, K. J. & Pines, D. (Eds.). (1988). *The Economy as an Evolving Complex System*. Redwood City, CA: Addison-Wesley.

Anthropic. (2026, February 27). Statement on the Comments from Secretary of War Pete Hegseth. https://www.anthropic.com/news/statement-comments-secretary-war

Arthur, W. B. (2021). Foundations of Complexity Economics. *Nature Reviews Physics*, 3(2). https://doi.org/10.1038/s42254-020-00273-3

AP News. (2025, January 28). Who is Liang Wenfeng, DeepSeek's founder? *AP News*. https://apnews.com/article/deepseek-founder-liang-wenfeng-china-ai-0673d5c39d90108189cc31b88d85b9f8

Barabási, A.-L. & Albert, R. (1999). Emergence of Scaling in Random Networks. *Science*, 286(5439). https://doi.org/10.1126/science.286.5439.509

Bass, T. A. (1999). *The Predictors: How a Band of Maverick Physicists Used Chaos Theory to Trade Their Way to a Fortune on Wall Street*. New York: Henry Holt.

Bloomberg. (2025, November 20). Robotics Startup Physical Intelligence Valued at $5.6 billion in new funding. *Bloomberg*. https://www.bloomberg.com/news/articles/2025-11-20/robotics-startup-physical-intelligence-valued-at-5-6-billion-in-new-funding

Bloomberg. (2026, February 6). Big Tech to Spend $650 Billion This Year as AI Race Intensifies. *Bloomberg*. https://www.bloomberg.com/news/articles/2026-02-06/how-much-is-big-tech-spending-on-ai-computing-a-staggering-650-billion-in-2026

Boyle, P. P. (1977). Options: A Monte Carlo Approach. *Journal of Financial Economics*, 4(3). https://doi.org/10.1016/0304-405X(77)90005-8

Caballero, R. J., Hoshi, T. & Kashyap, A. K. (2008). Zombie Lending and Depressed Restructuring in Japan. *American Economic Review*, 98(5). https://doi.org/10.1257/aer.98.5.1943

Chancellor, E. (2022). *The Price of Time: The Real Story of Interest*. New York: Atlantic Monthly Press.

DeepSeek. (n.d.). Organization card. Hugging Face. Retrieved October 14, 2026. https://huggingface.co/deepseek-ai

Federal Reserve Bank of St. Louis. (2026a). *Federal Funds Effective Rate* [FEDFUNDS]. FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/FEDFUNDS (Monthly series, 1954-07 to 2026-04, retrieved 2026-05-25)

Federal Reserve Bank of St. Louis. (2026b). *Total Assets for Bank of Japan* [JPNASSETS]. FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/JPNASSETS (Monthly series, units: 100 million yen, 1998-04 to 2026-04, retrieved 2026-05-25)

Feigenbaum, E. A. & McCorduck, P. (1983). *The Fifth Generation: Artificial Intelligence and Japan's Computer Challenge to the World*. Reading, MA: Addison-Wesley.

Figure AI. (2025, September 16). Figure Exceeds $1B in Series C Funding at $39B Post-Money Valuation. https://www.figure.ai/news/series-c

Finextra. (2005, November 15). UBS Buys Out Prediction Company. *Finextra*. https://www.finextra.com/newsarticle/14521/ubs-buys-out-prediction-company

Haigh, T. & Ceruzzi, P. E. (2021). *A New History of Modern Computing*. Cambridge, MA: MIT Press.

He, T. (2026). A Time Scaling Theory for Multi-Layer Electronic Systems. Keynote, *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*, Shanghai. Preprint: ChinaXiv 202605.00224v1. https://chinaxiv.org/abs/202605.00224v1

Hennessy, J. L. & Patterson, D. A. (2019). A New Golden Age for Computer Architecture. *Communications of the ACM*, 62(2). https://doi.org/10.1145/3282307

Hestness, J., Narang, S., Ardalani, N. et al. (2017). Deep Learning Scaling Is Predictable, Empirically. arXiv:1712.00409. https://arxiv.org/abs/1712.00409 [Preprint, not yet peer-reviewed]

High-Flyer. (n.d.). Company history. (In Chinese.) Retrieved October 14, 2026. https://www.high-flyer.cn/history/

Hinton, G. E., Osindero, S. & Teh, Y.-W. (2006). A Fast Learning Algorithm for Deep Belief Nets. *Neural Computation*, 18(7). https://doi.org/10.1162/neco.2006.18.7.1527

Kaplan, J., McCandlish, S., Henighan, T. et al. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361. https://arxiv.org/abs/2001.08361 [Preprint, not yet peer-reviewed]

Kenney, M., Han, K. & Tanaka, S. (2007). Venture Capital in Asia. In H. S. Rowen, M. G. Hancock & W. F. Miller (Eds.), *Making IT: The Rise of Asia in High Tech*. Stanford, CA: Stanford University Press.

Khandani, A. E. & Lo, A. W. (2011). What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data. *Journal of Financial Markets*, 14(1). https://doi.org/10.1016/j.finmar.2010.01.003

Kobayashi, K. & Inaba, M. (2002). Japan's Lost Decade and the Complexity Externality. *RIETI Discussion Paper Series*, 02-E-004. https://www.rieti.go.jp/en/publications/summary/02030004.html

Li, Q., Tai, C. & E, W. (2017). Stochastic Modified Equations and Adaptive Stochastic Gradient Algorithms. In *Proceedings of the 34th International Conference on Machine Learning* (ICML 2017). https://proceedings.mlr.press/v70/li17f.html

Li, Wei. (2022). Analyzing U.S. Semiconductor Industry Hegemony: A Political Economy Analysis of Industrial Power. *Foreign Affairs Review*, 2022(1). (In Chinese.) https://doi.org/10.13569/j.cnki.far.2022.01.022

Ma, Y. & Zimmermann, K. (2023). Monetary Policy and Innovation. *NBER Working Paper No. 31698*. https://doi.org/10.3386/w31698 [Working paper, not yet peer-reviewed]

Malinovsky, B. N. (2010). *Pioneers of Soviet Computing* (A. Fitzpatrick, Ed.). [Open electronic edition; original Russian edition 1995]. http://www.sigcis.org/files/SIGCISMC2010_001.pdf

Mandelbrot, B. B. & Hudson, R. L. (2004). *The (Mis)behavior of Markets: A Fractal View of Financial Turbulence*. New York: Basic Books.

Mandt, S., Hoffman, M. D. & Blei, D. M. (2017). Stochastic Gradient Descent as Approximate Bayesian Inference. *Journal of Machine Learning Research*, 18(134). https://jmlr.org/papers/v18/17-214.html

Markowitz, H. (1952). Portfolio Selection. *Journal of Finance*, 7(1). https://doi.org/10.2307/2975974

McCulloch, W. S. & Pitts, W. (1943). A Logical Calculus of the Ideas Immanent in Nervous Activity. *Bulletin of Mathematical Biophysics*, 5(4). https://doi.org/10.1007/BF02478259

Merton, R. C. (1969). Lifetime Portfolio Selection Under Uncertainty: The Continuous-Time Case. *Review of Economics and Statistics*, 51(3). https://doi.org/10.2307/1926560

MIT Technology Review. (2026, January 22). Yann LeCun's New Venture Is a Contrarian Bet Against Large Language Models. *MIT Technology Review*. https://www.technologyreview.com/2026/01/22/1131661/yann-lecuns-new-venture-ami-labs/

NRC (National Research Council). (1999). *Funding a Revolution: Government Support for Computing Research*. Washington, DC: National Academies Press. https://doi.org/10.17226/6323

OFweek. (2026, April). Two Financings per Day, 300 Million Entering the Field: Embodied Intelligence Stages a Capital Explosion. (Reproduced from Shuzhi Qianxian; annual financing data cited from ITJuzi.) (In Chinese.) [原文题：每天 2 起融资、3 亿入场，具身智能上演资本核爆] *OFweek*. https://m.ofweek.com/ai/2026-04/ART-201717-12005-30684477.html

Packard, N. H., Crutchfield, J. P., Farmer, J. D. et al. (1980). Geometry from a Time Series. *Physical Review Letters*, 45(9). https://doi.org/10.1103/PhysRevLett.45.712

Patterson, S. (2010). *The Quants: How a New Breed of Math Whizzes Conquered Wall Street and Nearly Destroyed It*. New York: Crown Business.

Patterson, S. (2012). *Dark Pools: The Rise of the Machine Traders and the Rigging of the U.S. Stock Market*. New York: Crown Business.

Peters, B. (2016). *How Not to Network a Nation: The Uneasy History of the Soviet Internet*. Cambridge, MA: MIT Press.

Russell, S. & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Hoboken, NJ: Pearson.

Song, Lei. (2016). *The Comparative Political Economy of Catch-up Industrial Strategies*. Beijing: The Commercial Press. (In Chinese.)

Stanford HAI (Institute for Human-Centered AI). (2026). *AI Index Report 2026*. Stanford University. https://hai.stanford.edu/ai-index/2026-ai-index-report

Tang, S. (2013). *The Social Evolution of International Politics*. Oxford: Oxford University Press.

TechCrunch. (2026a, March 9). Yann LeCun's AMI Labs Raises $1.03B to Build World Models. *TechCrunch*. https://techcrunch.com/2026/03/09/yann-lecuns-ami-labs-raises-1-03-billion-to-build-world-models/

TechCrunch. (2026b, March 27). Physical Intelligence Is Reportedly in Talks to Raise $1B, Again. *TechCrunch*. https://techcrunch.com/2026/03/27/physical-intelligence-is-reportedly-in-talks-to-raise-1-billion-again/

U.S. Department of War. (2025, December 22). The War Department to Expand AI Arsenal on GenAI.mil with xAI. https://www.war.gov/News/Releases/Release/Article/4366573/the-war-department-to-expand-ai-arsenal-on-genaimil-with-xai/

U.S. Department of War. (2026a, January 9). *Artificial Intelligence Strategy for the Department of War: Accelerating America's Military AI Dominance* [Memorandum]. https://media.defense.gov/2026/Jan/12/2003855671/-1/-1/0/ARTIFICIAL-INTELLIGENCE-STRATEGY-FOR-THE-DEPARTMENT-OF-WAR.PDF

U.S. Department of War. (2026b, May 1). Classified Networks AI Agreements. https://www.war.gov/News/Releases/Release/Article/4475177/classified-networks-ai-agreements/

Vaswani, A., Shazeer, N., Parmar, N. et al. (2017). Attention Is All You Need. In *Advances in Neural Information Processing Systems 30* (NeurIPS 2017). https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

Zuckerman, G. (2019). *The Man Who Solved the Market: How Jim Simons Launched the Quant Revolution*. New York: Portfolio/Penguin.